Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Seasonal Autoregressive Integrated Moving Average Model

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in/drsithuaung/

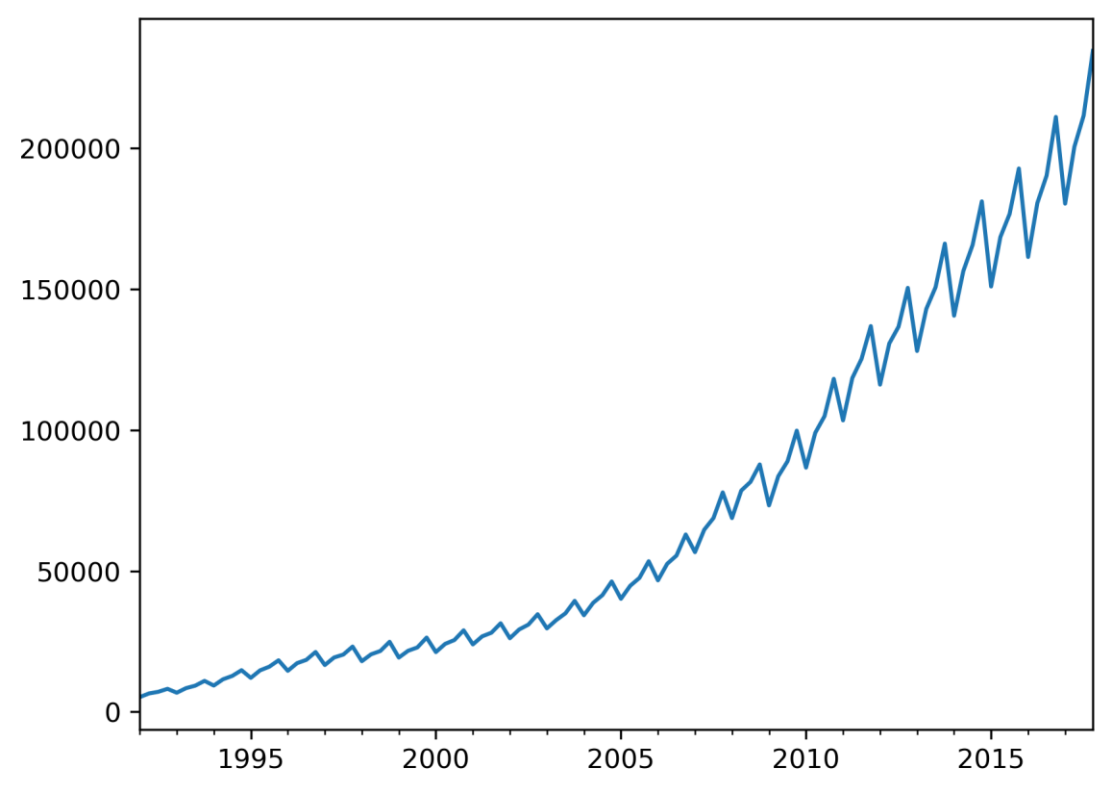
https://sithuaung-ink.github.io/data-analyst-portfolio/

3/25/2026

Case Study 1

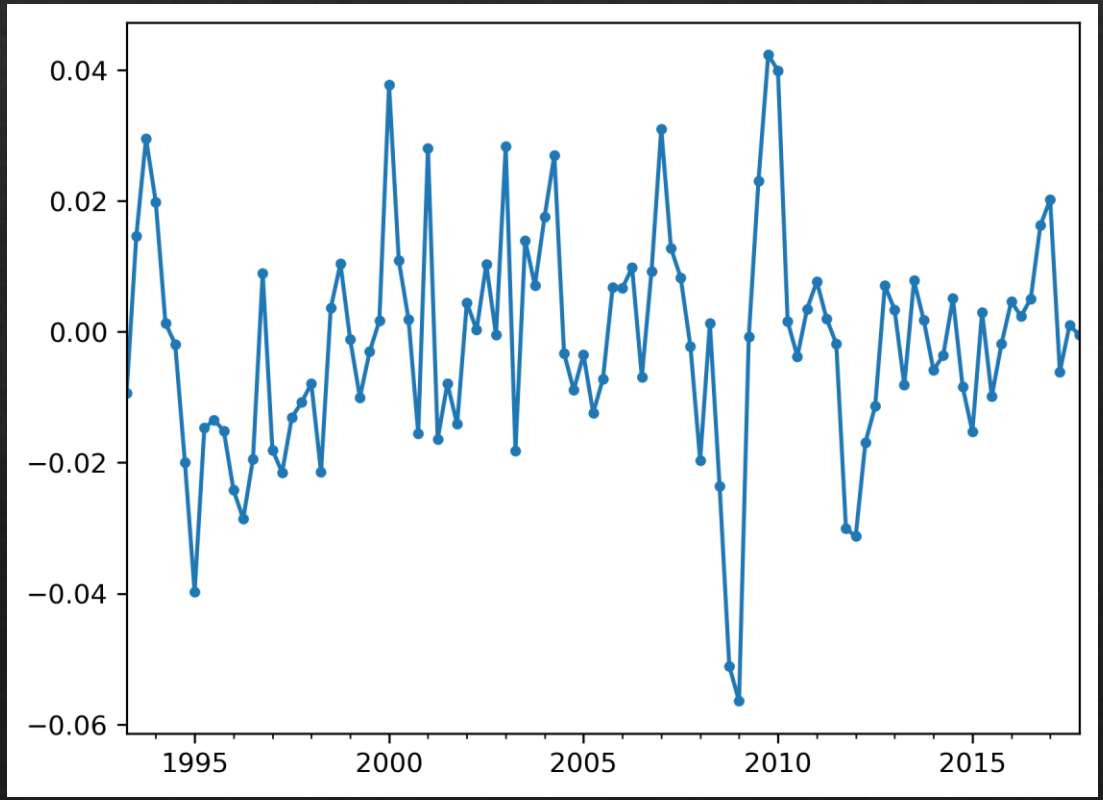
Chinese Quarterly GDP

- We will use the Chinese quarterly GDP from 1992 to 2017.



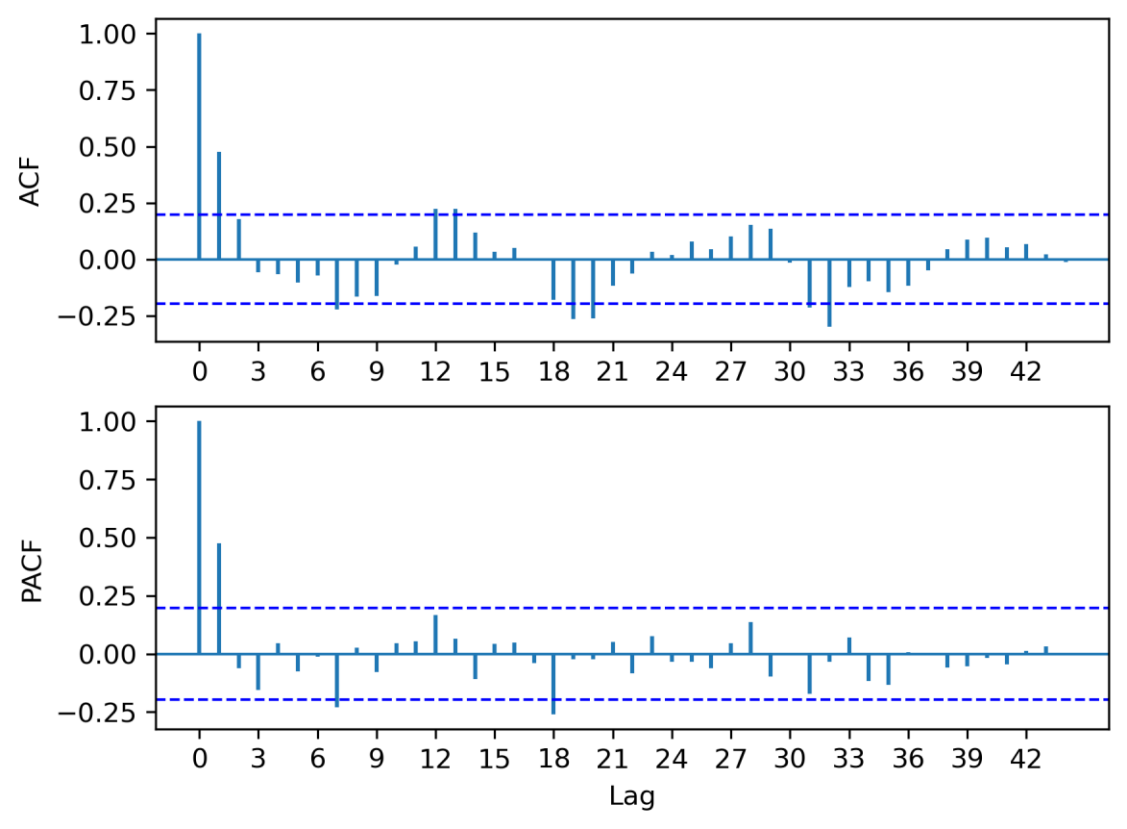
Case Study 1

Time series plot after seasonal and trend differenced



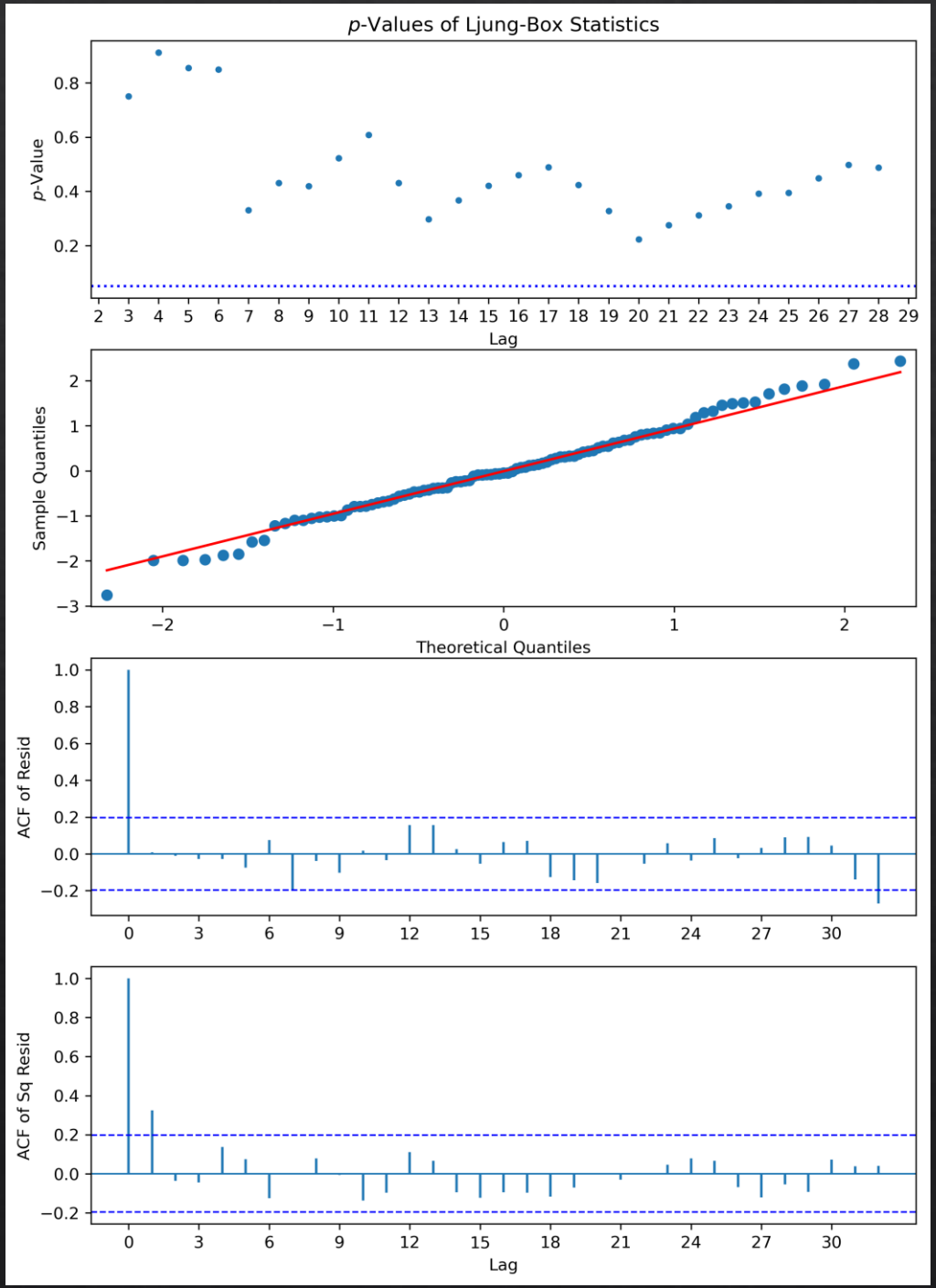
Case Study 1

ACF and PACF of the Chinese quarterly GDP



Case Study 1

Model diagnostic plots



Case Study 1

$$(1-B)(1-B^4)\ln(X_t) = (1+0.4940B+0.2688B^2)\varepsilon_t$$

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                   GDP   No. Observations:                  104
Model:             SARIMAX(0, 1, 2)x(0, 1, [], 4)   Log Likelihood                 274.805
Date:                            Wed, 25 Mar 2026   AIC                           -543.610
Time:                                    11:37:13   BIC                           -535.824
Sample:                                03-31-1992   HQIC                          -540.460
                                     - 12-31-2017
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1          0.4940      0.080      6.141      0.000       0.336       0.652
ma.L2          0.2688      0.102      2.644      0.008       0.070       0.468
sigma2         0.0002   3.19e-05      7.100      0.000       0.000       0.000
===================================================================================
Ljung-Box (L1) (Q):                   0.01   Jarque-Bera (JB):                 0.02
Prob(Q):                              0.92   Prob(JB):                         0.99
Heteroskedasticity (H):               0.52   Skew:                            -0.00
Prob(H) (two-sided):                  0.07   Kurtosis:                         3.07
===================================================================================
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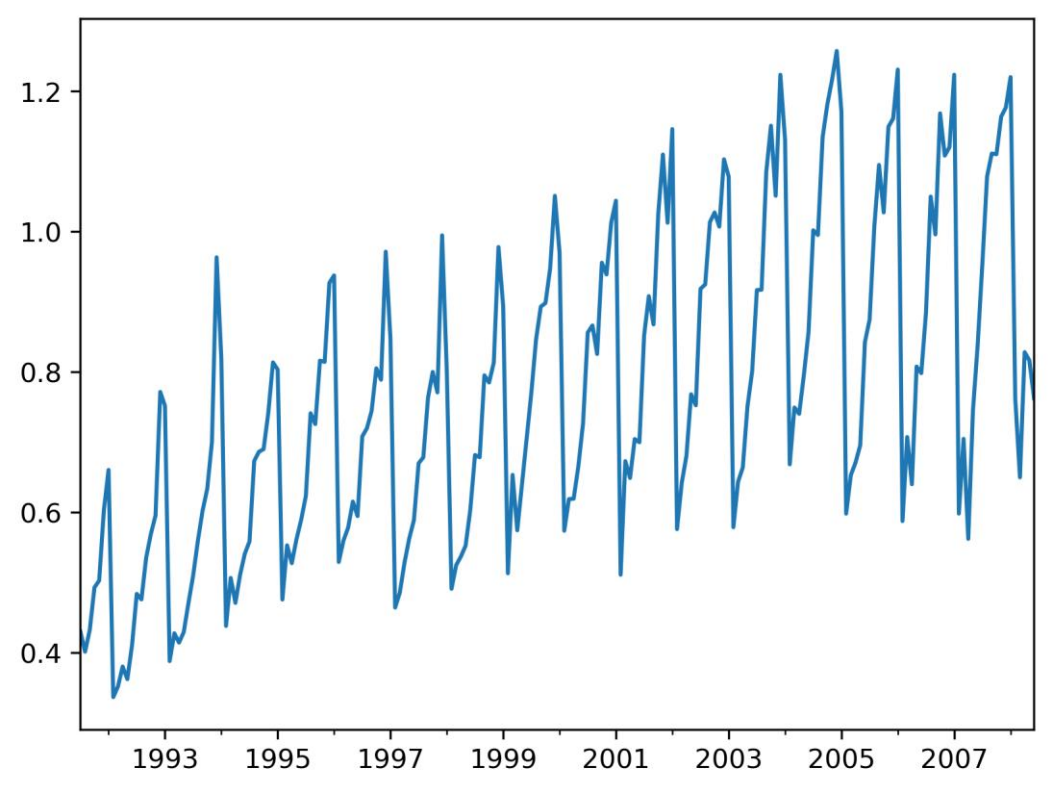
Case Study 2

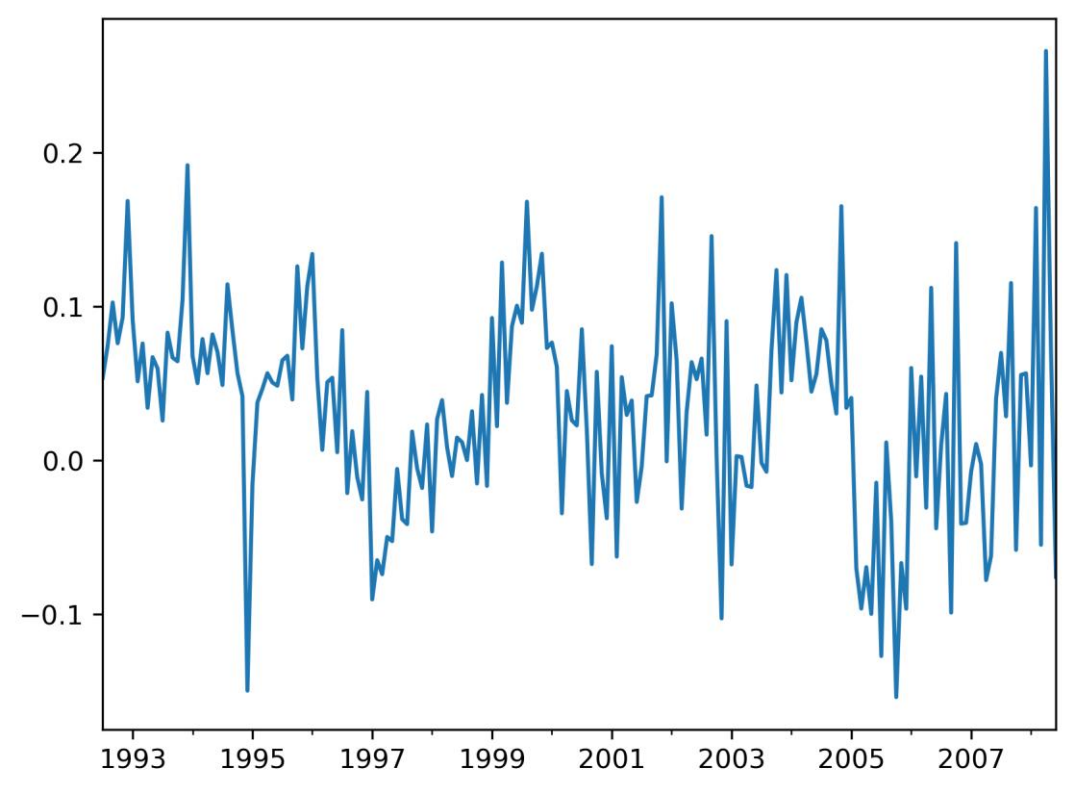
Corticosteroid Drug Sales in Australia

- We will use the Corticosteroid Drug Sales in Australia from July 1991 to June 2008.



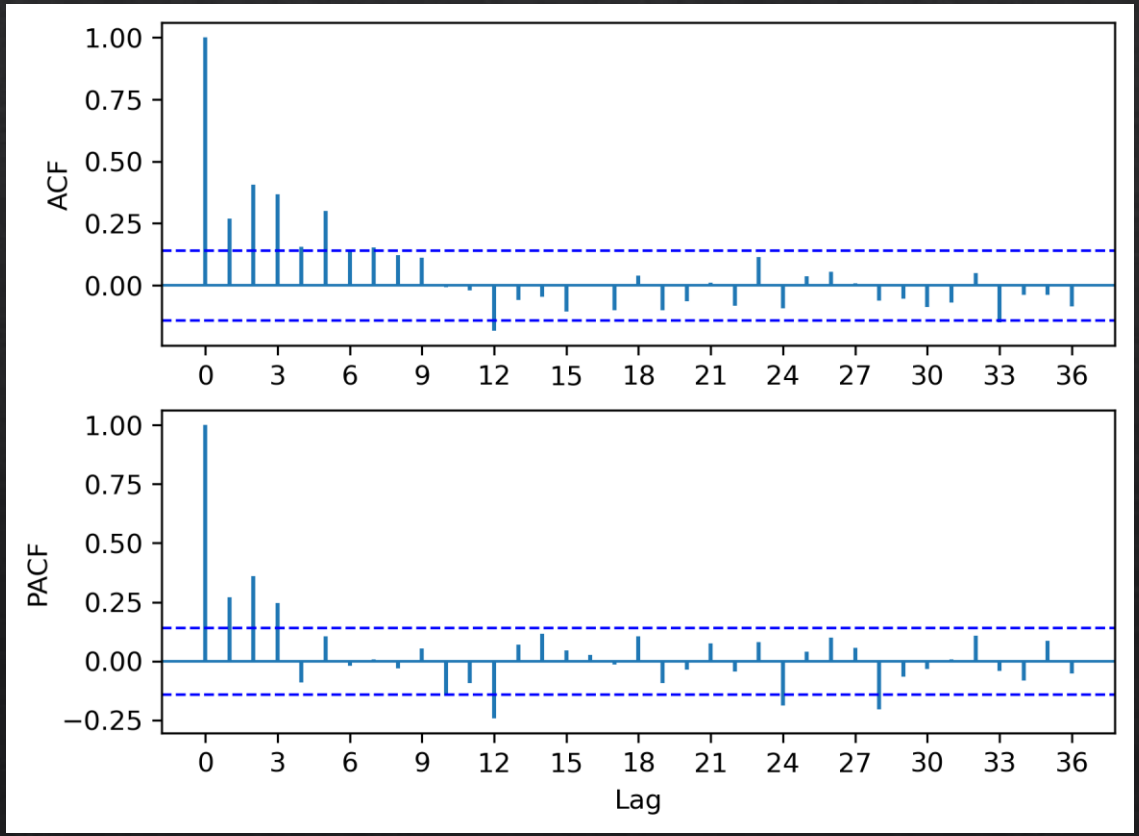
Case Study 2

Time series plot after seasonal differenced



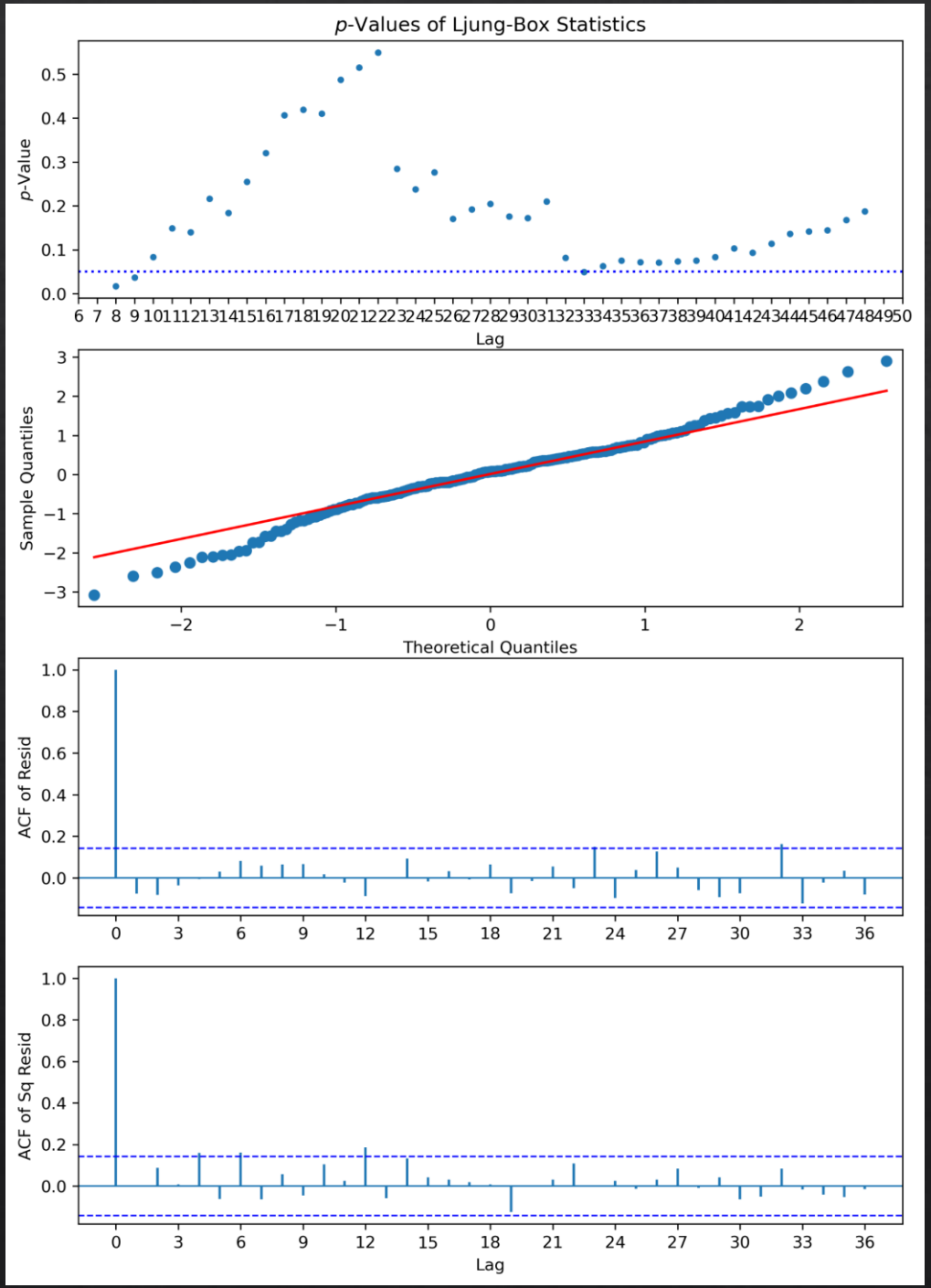
Case Study 2

ACF and PACF of the drug sales



Case Study 2

Model diagnostic plots

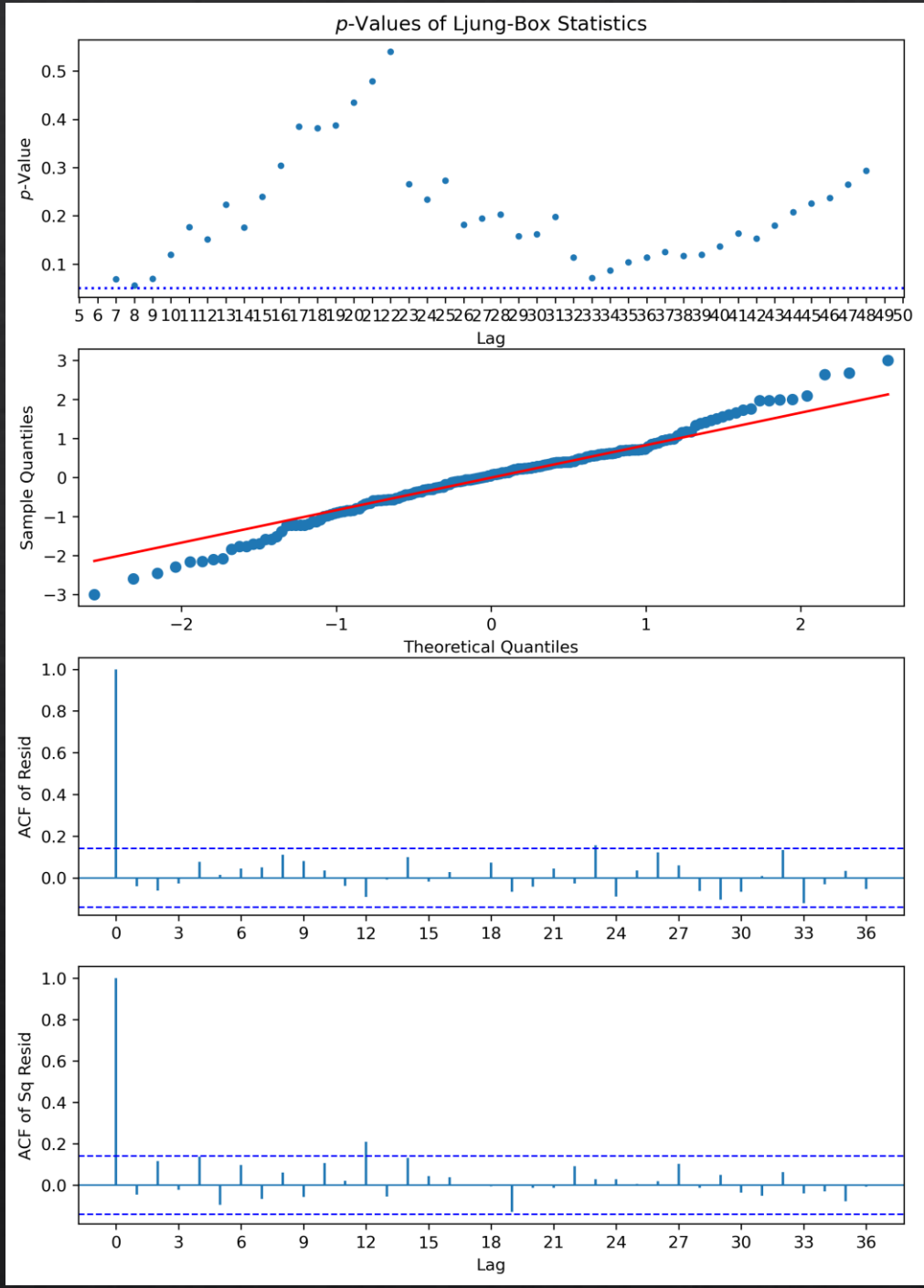


Case Study 2

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                  h02   No. Observations:                  204
Model:             SARIMAX(0, 0, 5)x(2, 1, [], 12)   Log Likelihood                 281.231
Date:                              Wed, 25 Mar 2026   AIC                           -546.462
Time:                                      11:50:35   BIC                           -520.402
Sample:                                  07-31-1991   HQIC                          -535.908
                                       - 06-30-2008
Covariance Type:                                opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1          0.2666      0.067      3.951      0.000       0.134       0.399
ma.L2          0.5501      0.072      7.599      0.000       0.408       0.692
ma.L3          0.5687      0.066      8.665      0.000       0.440       0.697
ma.L4          0.1406      0.076      1.847      0.065      -0.009       0.290
ma.L5          0.3591      0.076      4.700      0.000       0.209       0.509
ar.S.L12      -0.2861      0.071     -4.003      0.000      -0.426      -0.146
ar.S.L24      -0.2965      0.077     -3.853      0.000      -0.447      -0.146
sigma2         0.0031      0.000     10.067      0.000       0.002       0.004
===================================================================================
Ljung-Box (L1) (Q):                   1.31   Jarque-Bera (JB):                 5.13
Prob(Q):                              0.25   Prob(JB):                         0.08
Heteroskedasticity (H):               1.67   Skew:                            -0.21
Prob(H) (two-sided):                  0.04   Kurtosis:                         3.68
===================================================================================
```

Case Study 2

Model diagnostic plots



Case Study 2

$$(1 + 0.2718B^{12} + 0.2590B^{24})\,(1 - B^4)X_t = (1 + 0.2049B + 0.4979B^2 + 0.5101B^3 + 0.3495B^5)\varepsilon_t$$

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                   h02   No. Observations:                204
Model:             SARIMAX(0, 0, [1, 2, 3, 5])x(2, 1, [], 12)   Log Likelihood     279.791
Date:                                  Wed, 25 Mar 2026   AIC                    -545.583
Time:                                          11:51:02   BIC                    -522.780
Sample:                                      07-31-1991   HQIC                   -536.347
                                           - 06-30-2008
Covariance Type:                                    opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ma.L1          0.2049      0.061      3.347      0.001       0.085       0.325
ma.L2          0.4979      0.064      7.786      0.000       0.373       0.623
ma.L3          0.5101      0.069      7.353      0.000       0.374       0.646
ma.L5          0.3495      0.075      4.682      0.000       0.203       0.496
ar.S.L12      -0.2718      0.068     -3.997      0.000      -0.405      -0.139
ar.S.L24      -0.2590      0.074     -3.478      0.001      -0.405      -0.113
sigma2         0.0031      0.000     10.337      0.000       0.003       0.004
===================================================================================
Ljung-Box (L1) (Q):                   0.43   Jarque-Bera (JB):                 4.04
Prob(Q):                              0.51   Prob(JB):                         0.13
Heteroskedasticity (H):               1.54   Skew:                            -0.08
Prob(H) (two-sided):                  0.09   Kurtosis:                         3.69
===================================================================================
```

Conclusion

- Today's lecture, we learned two case studies of the SARIMA models.
- We learned how to choose the order of SARIMA models and seasonal periods.
- Ensure that the model's assumptions are validated by performing residual diagnostics.
- SARIMA models are a powerful tool in the analysis of time series since they are capable of modelling a wide range of time series.